Edge-Based Defense in Networks

A Modification of Bloch, Chatterjee & Dutta (2023)

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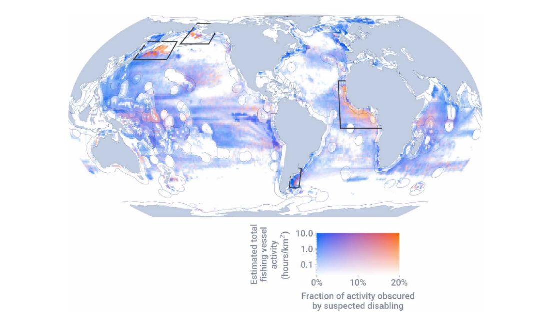
Strategic adversaries on networks



Estimated global fishing activity.
Orange bands mark suspected illegal activity — West Africa is one of three concern zones.

Welch et al., *Science Advances* (2022) — Global Fishing Watch

The problem.

Illegal fishing reaches places no single country owns. Vessels move along sea routes between national waters and high seas, taking what isn't theirs to take.

The adversary is strategic.

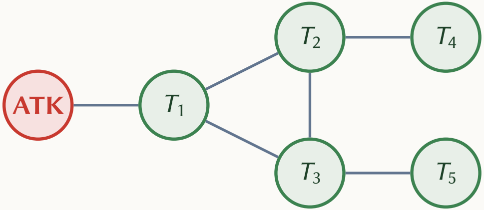
Routes are chosen, not random. The weakest gap is the one used.

The defenders are many.

Neighbouring states watch the same waters. Defense is naturally shared.

What is a network?

A network is a graph $G = (V, E)$: nodes V are locations, edges E are direct connections between them.



Nodes.

Locations the attacker can reach.

Edges.

The connections between locations.

The attacker.

A single adversary outside the network.

The defenders.

One at each target. They invest to intercept.

Two traditions, one gap

	Node Defense		Edge Defense
	Centralized	Continuous	Centralized
Authors	Dziubiński & Goyal Cerdeiro et al.	**Bloch, Chatterjee & Dutta (2023)**	Washburn & Wood Mavronicolas et al.
Investment	Binary	Continuous	Discrete or budgeted
Defenders	Single planner	Many	Single planner

The gap — *Continuous, decentralized investment — placed on the connections.*

The Bloch baseline

$$\alpha_{k(p)}(p) = \prod_{\substack{j \in V(p) \\ j \neq 0,\, j \neq k(p)}} (1 - x_j)$$

$$U(\pi, x) = \sum_{p \in P} \pi(p)\, \beta_{k(p)}(p)\, b_{k(p)}$$

$$x_i = 1 - \frac{U}{b_i}, \quad q_i = \frac{1}{b_i}\left(1 - \frac{U}{b_i}\right)$$

Path survival.

The attacker survives the interior of the path if it slips past every defender between node 0 and the target.

Attacker's expected payoff.

Sum over every path, weighted by how often the attacker uses it, times the chance of getting through, times the target's value.

Equilibrium for a first target.

U is the attacker's payoff. Higher-value targets invest more and are attacked more often.